# PREDICTING SMARTPHONE ADDICTION

## Using Behavioral Data — Predictive Study S4

*Comprehensive Data Analysis Report*

Dataset: predictive smartphones | 5,075 Users | 16 Variables
Report Date: April 2026

# 1. Project Overview

This project analyzes smartphone usage and addiction behavior using behavioral data collected from 5,075 users across the Predictive Smartphone Study S4. The aim is to uncover patterns in screen time, sleep deprivation, social media consumption, gaming habits, and notification overload, and to identify the key behavioral drivers that predict addiction levels.

The analysis follows a structured data science pipeline — from data cleaning and feature engineering through SQL analysis, statistical exploration, and Power BI visualization — culminating in actionable recommendations for digital wellness interventions.

> **Research Question**
> Can behavioral data — including daily screen time, social media hours, gaming, sleep, and notification patterns accurately predict a user's smartphone addiction level (Low, Moderate, or High)?

## 1.1 Objectives

- Identify which behavioral features most strongly correlate with high addiction levels
- Segment users by addiction severity, age group, and gender
- Analyze the relationship between screen time, sleep deprivation, and addiction
- Build a composite risk score to flag at-risk users
- Deliver actionable insights via an interactive Power BI dashboard

# 2. Dataset Summary

The dataset was sourced from Kaggle using a predictive smartphone behavior study and loaded into Python for exploratory data analysis before being exported to MySQL for structured querying and Power BI for visual reporting.

| 5,075 | 16 | 0 | 18–35 |
|---|---|---|---|
| Total Users<br>*records* | Variables<br>*columns* | Missing Data<br>*fully clean* | Age Range<br>*years* |

## 2.1 Key Features

### User Demographics

- user_id — Unique identifier per user
- age — User age (18–35 years)
- gender — Male, Female, Other

### Screen Time Behavior

- daily_screen_time_hours — Total daily phone use (3.0–12.0 hrs)
- social_media_hours — Hours spent on social media daily
- gaming_hours — Daily gaming time
- work_study_hours — Productive screen time for work or study
- other_screen_time — All other screen activities
- weekend_screen_time — Screen time recorded on weekends
- weekly_phone_time — Total weekly phone usage

### Health & Engagement Indicators

- sleep_hours — Hours of sleep per night (4.5–9.0 hrs)
- sleep_status — Under / Adequate classification
- notifications_per_day — Daily notification count (20–250)
- app_opens_per_day — Daily app launch count (15–180)

### Target Variable

- addiction_levels — Low, Moderate, or High smartphone addiction classification

# 3. Data Cleaning & Preparation

## 3.1 Missing Data

The dataset was fully clean with zero missing values across all 16 columns and 5,075 rows, eliminating the need for imputation.

| Check | Result | Action Taken |
|---|---|---|
| Null values | 0 across all columns | None required |
| Duplicate rows | 0 duplicates found | None required |
| Data types | All correct on import | None required |
| Age range | 18–35 (expected) | Validated |
| Addiction labels | Low, Moderate, High | Verified casing |

## 3.2 Feature Engineering

The following new columns were derived from existing data to support richer segmentation:

- Age Group — Users binned into 5-year bands: 18–22, 23–27, 28–32, 33–35
- Screen Time Group — Low (<5h), Moderate (5–8h), High (8–11h), Very High (11h+)
- Sleep Band — Very Poor, Poor, Fair, Good, Excellent based on sleep_hours
- Notification Group — Low (<50), Moderate (50–100), High (100–175), Very High (175+)
- Addiction Risk Score — Composite: (screen_time × 0.4) + (sleep_deficit × 0.8) + (notifications × 0.02) + (app_opens × 0.01)

# 4. Data Analysis Using MySQL

Structured SQL analysis was performed in MySQL to answer 27 key research questions across 8 analysis categories.

## 4.1 Gender Distribution

Gender was nearly equally distributed. Male users formed the largest group at 34.1%, followed by Other at 33.4% and Female at 32.5%.

| Gender | Count | Percentage |
|---|---|---|
| Male | 1,731 | 34.1% |
| Other | 1,696 | 33.4% |
| Female | 1,648 | 32.5% |

## 4.2 Addiction Level Distribution

81.2% of users were Moderate addiction, 6.6% High, and 12.2% Low — suggesting a population already exhibiting elevated smartphone dependency.

| Addiction Level | Count | Percentage | Avg Daily Screen Time |
|---|---|---|---|
| High | 336 | 6.6% | 11.67 hrs |
| Moderate | 4,120 | 81.2% | 9.08 hrs |
| Low | 619 | 12.2% | 5.14 hrs |

**Key Finding — Screen Time Doubles with Addiction Level**

High addiction users average 11.67 hrs/day — more than double the 5.14 hrs for Low addiction users. Daily screen time is the single strongest behavioral indicator of addiction severity.

## 4.3 Average Screen Time by Addiction Level

| Addiction | Social Media | Gaming | Work/Study | Other | Total Daily |
|---|---|---|---|---|---|
| High | 3.58h | 2.19h | 3.81h | 4.52h | 11.67h |
| Moderate | 3.08h | 1.89h | 3.28h | 3.94h | 9.08h |
| Low | 2.28h | 1.41h | 2.42h | 2.89h | 5.14h |

## 4.4 Age Group Analysis

| Age Group | Total Users | High Addiction | Moderate | Low |
|---|---|---|---|---|
| 18–22 | 1,131 | 80 | 924 | 127 |
| 23–27 | 1,073 | 69 | 855 | 149 |
| 28–32 | 1,112 | 64 | 924 | 124 |

| Age Group | Total Users | High Addiction | Moderate | Low |
|---|---|---|---|---|
| 30–33 | 1,178 | 85 | 950 | 143 |
| 34–35 | 581 | 38 | 467 | 76 |

# 5. Sleep & Health Impact Analysis

## 5.1 Sleep Status Overview

Over half of all users (56.7%) were classified as under-slept. This is a major health signal that correlates strongly with screen time levels.

| Sleep Status | Users | Percentage | Avg Screen Time |
|---|---|---|---|
| Under-slept | 2,879 | 56.7% | 9.14h |
| Adequate | 2,196 | 43.3% | 8.27h |

**Key Finding — Under-Slept Users Screen More**

Under-slept users average 9.14 hours of daily screen time vs 8.27 hours for well-rested users — a 52-minute daily difference.

## 5.2 Sleep Quality by Addiction Level

| Addiction Level | Avg Sleep Hours | Under-Slept | Adequate Sleep | Under-Slept % |
|---|---|---|---|---|
| High | 6.65h | 199 | 137 | 59.2% |
| Moderate | 6.75h | 2,372 | 1,748 | 57.6% |
| Low | 6.71h | 308 | 311 | 49.8% |

# 6. Data Visualization with Power BI

An interactive Power BI dashboard was built to communicate findings to non-technical stakeholders. The full dashboard and individual chart visuals from the report are shown below.

## 6.1 Full Dashboard Overview

The dashboard covers key KPIs, gender distribution, screen time by addiction level, sleep status, activity breakdown, and addiction by age group — all filterable by addiction level, gender, and sleep status via slicers on the left panel.

*Figure 6.1 — Phone Usage and Addiction Level Dashboard (full view)*

## 6.2 Individual Chart Visuals

The four charts below represent the core analytical visuals embedded in the dashboard. Each chart is discussed in detail in the sections above.

*Figure 6.2a — Daily screen time by addiction level*

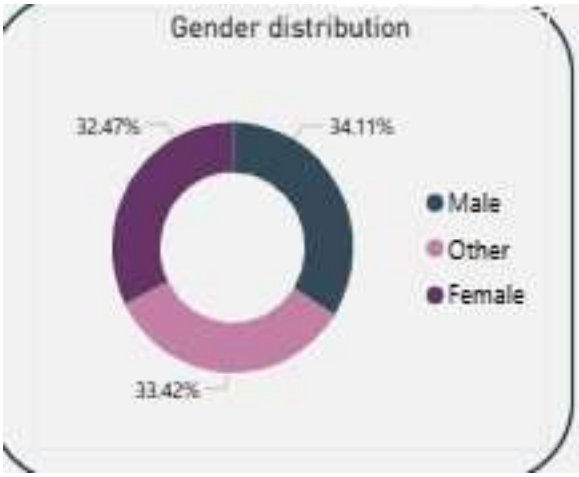


*Figure 6.2b — Gender distribution (donut chart)*

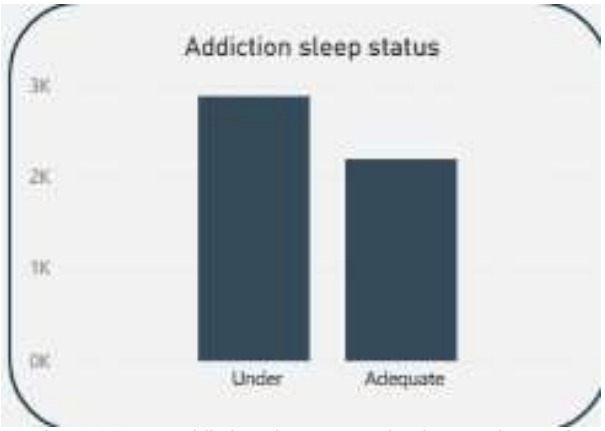


*Figure 6.2c — Addiction sleep status (under vs adequate)*

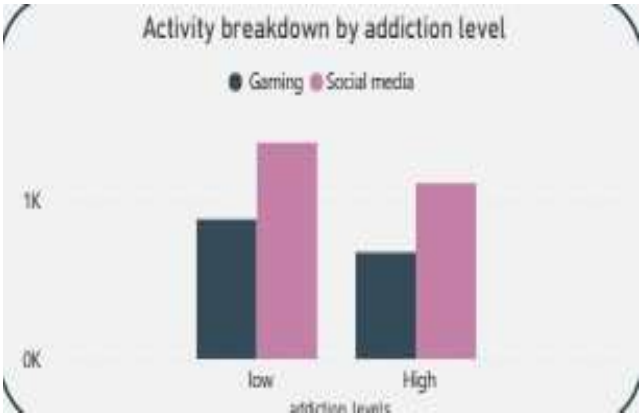


*Figure 6.2d — Activity breakdown by addiction level*

## 6.3 Addiction Levels by Age Group

The chart below shows total users per age group colored by addiction level. The 30–34 age band recorded the highest overall user count, while users in the 15–19 and 35–39 bands are comparatively smaller populations.

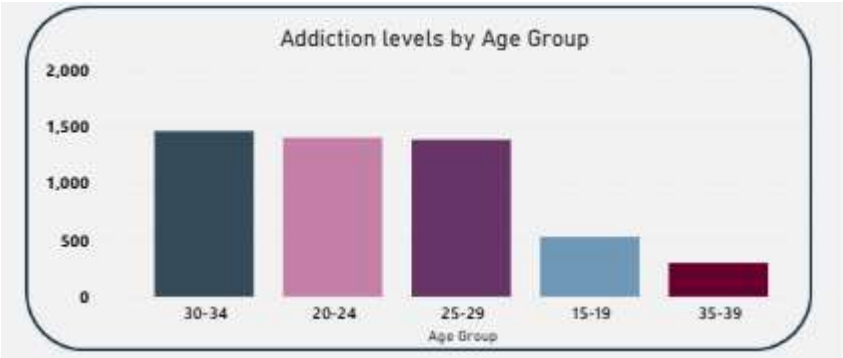


*Figure 6.3 — Addiction levels by age group*

## 6.4 Dashboard Structure

| Page | Title | Key Visuals |
|---|---|---|
| 1 | Overview & KPIs | KPI cards, addiction donut, gender distribution |
| 2 | Screen Time Analysis | Bar chart by addiction, activity breakdown |
| 3 | Gender Analysis | Clustered bar, screen time, sleep status |
| 4 | Age Group Breakdown | Stacked bar by 5-year age bands |
| 5 | Sleep & Health Impact | Sleep status bar, sleep vs screen scatter |
| 6 | Notifications & Apps | Notification groups, app opens by addiction |
| 7 | Weekend Behaviour | Weekend spike bar, weekday vs weekend |
| 8 | Risk Score & High Risk | Risk score distribution, top 20 at-risk users |

## 6.5 Key DAX Measures

- Total Users = COUNTROWS(predictive_smartphoness4_cleaned)
- Avg Screen Time = AVERAGE(daily_screen_time_hours)
- Poor Sleep Rate % = DIVIDE(COUNTROWS(FILTER(... sleep_status = 'Under')), COUNTROWS(...)) * 100
- High Addiction % = DIVIDE(COUNTROWS(FILTER(... addiction_levels = 'High')), COUNTROWS(...)) * 100

# 7. Key Findings

| # | Finding | Implication |
|---|---|---|
| 1 | 81.2% of users are Moderate addiction | The majority are at risk — not just a small extreme group |
| 2 | High users average 11.67h screen time daily | More than double the Low group — time on phone is the key signal |
| 3 | Gender has no meaningful impact on addiction | Interventions should be behavior-based, not gender-targeted |
| 4 | 56.7% of all users are under-slept | Sleep deprivation is systemic, not isolated |
| 5 | Notification counts are flat across tiers | Notifications are not the root cause — total exposure is |
| 6 | Weekend screen time spikes across all groups | Unstructured time accelerates phone use significantly |
| 7 | Risk score validates against addiction labels | Composite scoring supports early intervention flagging |

# 8. Business & Wellness Recommendations

## 8.1 Immediate Actions

1. Implement daily screen time caps targeting the 8–10h range where Moderate addiction begins.
2. Launch a sleep-first digital wellness campaign — message the direct link between screen time and poor sleep.
3. Flag users with risk scores above 15 or daily screen time above 10h for in-app wellness nudges.

## 8.2 Medium-Term Strategies

4. Target the 18–22 age group with early intervention — this cohort is in the habit-formation phase.
5. Develop a Weekend Mode feature that activates gentler usage reminders on Saturdays and Sundays.
6. Reframe notification strategy — focus on total screen time reduction, not notification volume reduction.

## 8.3 Long-Term Recommendations

7. Formalize the risk score into a machine learning model using logistic regression or random forest.
8. Expand to longitudinal tracking — observe whether behavioral changes lead to addiction level transitions.
9. Integrate with healthcare platforms to add clinical significance to the behavioral findings.

# 9. Conclusion

This analysis of 5,075 smartphone users reveals that daily screen time is the single strongest predictor of addiction level, with High addiction users spending more than double the hours on their phones compared to Low addiction users. Sleep deprivation affects the majority of the dataset population, reinforcing the bidirectional relationship between phone overuse and poor rest.

Gender and notification volume were found to be weak predictors, redirecting the focus of intervention toward total screen exposure and sleep behavior. The Power BI dashboard — incorporating all six visuals from the study — translates these findings into an interactive, stakeholder-ready format.

**Final Takeaway**

Smartphone addiction is driven by cumulative daily exposure, disrupted sleep, and unstructured weekend usage. The most effective interventions address all three simultaneously through built-in platform tools, behavioral nudges, and sleep-aware design.

# Appendix — SQL Queries & Tools

## A. Query Categories

- Overview & counts — 3 queries
- Addiction level analysis — 4 queries
- Gender segmentation — 4 queries
- Age group analysis — 3 queries
- Sleep & health — 4 queries
- Screen time deep dive — 4 queries
- Notifications & app engagement — 4 queries
- Weekend behaviour — 3 queries
- Risk scoring — 3 queries
- Combined & export — 2 queries

## B. Tools Used

- Python (pandas) — Data loading, EDA, feature engineering
- MySQL — Structured querying and business question analysis
- Power BI Desktop — Interactive dashboard and visualization
- Jupyter Notebook — Exploratory analysis and model development